Binomial theorem for a positive integral index:

For any natural number n

$$(x+a)^n = nc_0 x^n a^0 + nc_1 x^{n-1} a^1 + nc_2 x^{n-2} a^2 + \ldots + nc_r x^{n-r} a^r + \ldots + nc_n a^n$$

In the above binomial expansion, we observe,

(i) The $(r+1)^{th}$ term is denoted by $T_{r+1} = nc_r x^{n-r} a^r$

(ii) The degree of 'x' in each term decreases while that of 'a' increases such that the sum of the power in each term equal to n

(iii) nc_0, nc_1, nc_2, … nc_r, … nc_n are binomial coefficients they are also written as $c_0, c_1, c_2 \ldots, c_n$

(iv) From the relation $nc_r = nc_{n-r}$ we see that the coefficients of term equidistant from the beginning and the end are equal.

Example 7.15

Expand $(2x+y)^5$ using binomial theorem.

Solution:

$$(x+a)^n = nc_0 x^n a^0 + nc_1 x^n - 1_a + nc_2 x^{n-2} a^2 + \ldots + nc_r x^{n-r} a^r + \ldots + nc_n a^n$$

Here, $n = 5$, $X = 2x$, $a = y$

$$\begin{aligned}(2x+y)^5 &= 5C_0 (2x)^5 (y)^0 + 5C_1 (2x)^4 (y)^1 + 5C_2 (2x)^3 (y)^2 + \\ &\quad 5C_3 (2x)^2 (y)^3 + 5C_4 (2x)^1 (y)^4 + 5C_5 (2x)^0 (y)^5\end{aligned}$$

$$= (1)2^5x^5 + (5)2^4x^4y + (10)2^3x^3y^2 + (10)2^2x^2y^3 + (5)2^1x^1y^4 + (1)y^5$$

$$= 32x^5 + 80x^4y + 80x^3y^2 + 40x^2y^3 + 10xy^4 + y^5$$

$$(2x+y)^5 = 32x^5 + 80x^4y + 80x^3y^2 + 40x^2y^3 + 10xy^4 + y^5$$

Example 7.16

Find the middle terms of expansion $(3x+y)^5$

Solution:

In the expansion of $(3x+y)^5$ we have totally 6 terms. From this the middle terms are

T_3 and T_4